BATTERY INPUTS / SERIES LINK + RESERVOIR
J4
12V BATTERY / UPPER
J30
12V BATTERY / LOWER
VBAT_24V
BAT_MID_12V
8 x 22uF / 50V / 2220 reservoir
C335
C336
C337
C338
C339
C340
C341
C342
22uF 50V X7R
Local rail bypass
C333
22uF 50V
24 V TO 12 V / DRIVER BIAS
R300
34k
R301
2.43k
C300
4.7nF
C301
22nF
R305
10k
24 V TO 5 V / LOGIC
C312
C313
U31
P31_SS
C310
4.7nF
R310
34k
R311
6.49k
C311
22nF
ANALOG + LOGIC SUPPLIES
+12V_BIAS
R740
10R
+12V_AUX
U2
AP2112K-3.3
C741
10uF 16V X7R
-5V_CT
SHARED −5 V / CHARGE PUMP
C420
1uF 16V
C421
1uF 16V
C422
10uF 16V
C423
10uF 16V
R421
10R
CT1_FLY_N
CT1_NEG_PUMP
PWR_FLAG
MOUNTING
H1
M3 module mounting
H2
M3
H3
M3
H4
M3
H5
M3
DAUGHTERBOARD / FPGA + ESP32
USB_VBUS_PRESENT
USB−C / PROGRAMMING ONLY
R24
5.1k
R25
5.1k
USB_CC1
USB_CC2
SHIELD
GND
DESIGN NOTES
AUX 1 / UNITY BUFFER
AUX 2 / UNITY BUFFER
AUX 3 / UNITY BUFFER
AUX 4 / UNITY BUFFER
+5V_ADC
D70
BAV199,215
D71
BAV199,215
D72
BAV199,215
C700
150pF 50V C0G
R701
1M 1%
C701
2.2nF 50V C0G
C703
100nF 16V X7R
U70
OPA197IDBVR
TP702
GND
TP711
AUX2_BUFFERED
TP712
GND
TP731
AUX4_BUFFERED
TP732
GND
ADC / AD7606C−16
U1
AD7606C−16BSTZ
TP1
ADC_FIRST
ANALOG INPUTS / ADC RC FILTERS
J3
AUX 1 / 0−5V
J8
AUX 2 / 0−5V
J9
AUX 3 / 0−5V
J10
AUX 4 / 0−5V
R1
R2
R3
R4
R5
R6
R7
R8
100R
C11
C12
C13
C14
C15
C16
C17
C18
1nF
ADC_CH1
ADC_CH2
ADC_CH3
ADC_CH4
ADC_CH5
ADC_CH6
ADC_CH7
ADC_CH8
ADC / LOCAL BYPASS + REFERENCE
VDRIVE
C1
C2
C3
C4
100nF
C10
10uF
C5
100nF
C6
1uF
C7
1uF
C8
100nF
C9
10uF
REGCAP_A
REGCAP_D
ADC_REF_2V5
ADC_REFCAP
LOCAL POWER + BRIDGE LEG INDICATORS
GDT2 L / GREEN
GDT2 R / GREEN
R807
12k
R808
12k
D87
GREEN
D88
GREEN
D92
1N4148W,115
D93
1N4148W,115
GDT 1 / FPGA COMMAND DAMPING
R500
2.2k
R501
2.2k
R506
33R
GDT1_L_LO
GDT1_L_LO_BUS
GDT 1 / FULL BRIDGE + GATE BIAS
U51
UCC27282DR
Q50
STD80N10F7
R529
15R 1W
C530
1uF 50V X7R
GDT1_SNUB
GDT1_OUT_P
8 x 22uF / 50V X7R / 2220 reservoir
C526
C531
C532
C533
C534
C535
C536
C537
22uF 50V X7R
D50
SMBJ30A
GDT 2 / FPGA COMMAND DAMPING
R513
33R
R515
33R
GDT2_L_HI
GDT2_L_HI_BUS
GDT2_R_HI
GDT2_R_HI_BUS
GDT 2 / FULL BRIDGE + GATE BIAS
CST 1 / INPUT + PRECISION RECTIFIER
CST 2 / INPUT + PRECISION RECTIFIER
C427
150pF 50V C0G
C428
150pF 50V C0G
C429
1nF 50V C0G
R405
10k 0.1%
R406
10k 0.1%
CT1_ABS_SUM
TP40
CT1_RAW
TP41
CT1_HALF
TP42
CT1_ABS
TP43
CT1_ENVELOPE
CST 1 / PHASE COMPARATOR + REFERENCE
CST 2 / PHASE COMPARATOR + REFERENCE
U42
TLV3501AIDBVR
C425
1uF 16V
R410
5k 0.1%
R411
10k 0.1%
R416
33R
D43
BAT54S,215
CT1_PHASE
CT1_MID
R415
1k 0.1%
C412
1uF
CST 1 / LOCAL BYPASS
CST 2 / LOCAL BYPASS
U40/U41 positive supply
Phase comparator supply bypass
C413
100nF
C424
2.2uF
+3V3

BATTERY INPUTS / SERIES LINK + RESERVOIR

Series: J4 upper battery, J30 lower battery; pin 1 +, pin 2 −.

Single 24 V PSU: + to J4.1, − to J30.2; leave J4.2 / J30.1 unused.

BAT_MID_12V joins J4.2 to J30.1 only; it is not board ground.

24 V TO 12 V / DRIVER BIAS

LMZ14202H: shielded inductor inside the module.

UVLO: approx. 21.0 V on / 19.4 V off; soft start 0.47 ms.

Output bank: verify DC−biased capacitance before release.

2 A device rating; thermal / load qualification pending.

24 V TO 5 V / LOGIC

LMZ14202H: shielded inductor inside the module.

UVLO: approx. 21.0 V on / 19.4 V off; soft start 0.47 ms.

Output bank: verify DC−biased capacitance before release.

2 A device rating; thermal / load qualification pending.

SHARED −5 V / CHARGE PUMP

250 kHz charge pump; 10R/10uF ripple filter.

Negative rail feeds both rectifiers and the AUX buffers.

USB−C / PROGRAMMING ONLY

PROGRAMMING ONLY − power the controller from its normal supply.

USB VBUS is isolated from VIN_5V. No USB−PD or USB−UART bridge.

VBUS presence: J1.20 / ESP GPIO8; firmware must monitor this input.

A6/B6 = D+; A7/B7 = D−. Each CC pin has its own 5.1k pulldown.

DESIGN NOTES

Input wiring: see BATTERY INPUTS / SERIES LINK + RESERVOIR.

Values and power−stage thermal performance require validation.

AUX 1 / UNITY BUFFER

1k / 2.2nF: 72 kHz nominal; ADC−side 100R / 1nF follows.

1M DC load; shared return; powered operation only.

ADC / AD7606C−16

OS[2:0]=111: software mode

Configure ranges and high bandwidth before capture.

ANALOG INPUTS / ADC RC FILTERS

CH1−4: AUX1, AUX2, AUX3, AUX4 / 0−5 V; CH5/6 CST 1; CH7/8 CST 2

100R / 1nF: EMI filtering, not anti−alias filtering.

ADC / LOCAL BYPASS + REFERENCE

AVCC / one local capacitor at each supply pin

REGCAP_A and REGCAP_D are internal−regulator outputs: bypass only.

REFCAP pins 44/45 share C9; REF pin 42 uses C8.

LOCAL POWER + BRIDGE LEG INDICATORS

Each bridge lamp senses its switching node through 12k; no FPGA status pins.

1N4148W protects each LED from negative switching−node excursions.

Brightness follows positive duty cycle; not a fault, enable or current monitor.

GDT 1 / FPGA COMMAND DAMPING

FPGA owns enable, zero−current shutdown, deadtime and flux balance.

GDT 1 / FULL BRIDGE + GATE BIAS

Precharge: both lows ON, both highs OFF. Insert deadtime before each leg commutation.

UD2.7: 2.2uF / 50V X7R DC block, with parallel 15R + 1uF / 50V X7R damping branch.

CST 1 / INPUT + PRECISION RECTIFIER

10R burden: 2 x 20R / nominal +/−2 V. Choose CT ratio N = 5 x primary peak amperes.

Vhalf = −max(Vct,0); U41 sums −(Vct + 2 Vhalf) = |Vct|. 1k/10nF: 10 us smoothing, not peak hold.

Keep a secondary clamp at the CT itself if the cable can unplug. Overcurrent decisions are made by the FPGA from ADC samples.

CST 1 / PHASE COMPARATOR + REFERENCE

Vphase = 1.65 V + Vct/2; compare with CT1_MID.

22pF EMI filter; 6mV internal comparator hysteresis.

Timing threshold + delay must be calibrated versus amplitude.

Phase midpoint: 1.65 V. Wait >=100 ms after rails stabilize before arming.

Sheet: /
File: controller.kicad_sch

Title: Tesla coil controller / sensing, control and power

Size: User	Date:	**Rev: A0 — NOT FOR FABRICAT**
KiCad E.D.A. 10.0.6		Id: 1/1